Group Assignment

Data Story Proposal

MBAI 5400G: Visualization & Storytelling Winter 2026

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Title

Estimating AI Exposure in Oshawa Workforce and Its Impact on Housing Affordability for Middle-Aged Workers.

Research Background

Artificial intelligence is increasingly affecting the workplace. Some occupations, for example administrative works, data entry, customer service, and business support roles can be more exposed to AI than others.

Not only these, in recent times housing prices in Oshawa have increased significantly. This creates a serious concern for middle-aged workers. If their jobs become unstable or require retraining, they will struggle to keep up with rising housing costs.

The main idea of this research is that some occupations in Oshawa can be more exposed to AI than others, and workers in those occupations will face employment uncertainty, wage pressure, or the need for retraining. If housing prices continue to rise while workers face these challenges, affordability will become more difficult. Our possible analytical flow is:

Workforce structure in Oshawa
↓
AI exposure across occupations
↓
Estimated number of workers affected by AI
↓
Housing price growth in Oshawa
↓
Overall workforce AI risk and affordability pressure

Grounding Question

- When AI Changes the Job Market, Can Middle-Aged Workers in Oshawa Still Afford a Home?

Possible Research Questions

- To what extent is the Oshawa workforce exposed to artificial intelligence?
- Which occupations in Oshawa are most vulnerable to AI-related disruption?

- How AI-driven labor market changes create affordability pressure for middle-aged workers who need to upskill or reskill?
- Will increasing AI exposure combined with rising housing prices create long-term financial instability for mid-career workers?

Possible Themes

- Artificial intelligence and labor market change
- Workforce reskilling and career transition
- Housing affordability
- Economic vulnerability of middle-aged workers
- Technology and local economic stability
- Future of work in local communities

Possible Characters

- Middle-aged workers in Oshawa who must upskill or reskill due to AI-related changes in their occupations.

Possible Data

- Statistics Canada – 2021 Census Profile (Occupation data for Oshawa)
- Stanford University – "GPTs are GPTs: An Early Look at the Labor Market Impact Potential of Large Language Models" dataset
- Statistics Canada – New Housing Price Index (NHPI)
- Statistics Canada – Labor Force characteristics by occupation

Possible Research Method

We will collect occupation data for Oshawa from Statistics Canada and AI exposure scores from the Stanford dataset. Then we will merge the two datasets by occupation and estimate how many workers will be exposed to AI by multiplying the number of workers by the AI exposure score. For example, if an occupation has 12,000 workers and an AI exposure score of 0.75, then about 9,000 workers will be AI-exposed. After this, we will calculate the overall AI exposure rate (Workforce AI Risk Index) by dividing the total AI-exposed workers by the total number of workers. For instance, if the result is 0.48, it means about 48% of jobs in Oshawa can be affected by AI to some degree. Finally, we will compare this result with housing price trends in Oshawa using a simple line chart to see whether AI-related job changes and rising housing costs together can create affordability pressure for middle-aged workers who need to reskill.

Possible Visual Methods

- Horizontal Bar Chart: Occupation vs Number of Workers
  - Shows the structure and size of different occupation groups in Oshawa.

- Bubble Chart: AI Exposure vs Number of Workers
  - Shows the relationship between AI exposure and workforce size, helping identify occupations with both high exposure and many workers.

- Stacked Bar Chart: AI-Exposed Workers by Occupation
  - Shows estimated AI-exposed workers compared with total workers in each occupation.

- Line Chart: Year vs Oshawa Housing Price
  - Shows housing price trends over time and highlights increasing affordability pressure.

- Indicator / Gauge Chart: Workforce AI Risk Index
  - Displays the overall proportion of Oshawa jobs potentially exposed to AI as a single summary measure.

Possible Formats

- Data-driven scrolly-telling method explaining AI exposure and affordability
- Interactive dashboard connecting workforce risk and housing trends

Possible Audience

- Our data driven story will help both policymakers and individual workers understand how AI-driven job changes and rising housing costs will affect financial stability in Oshawa.